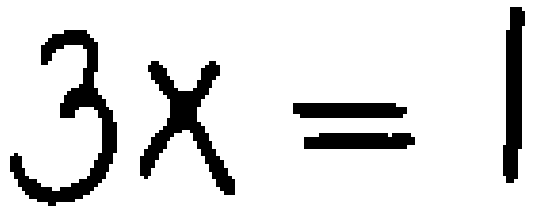
lecture_01_10423__formula

ACTUAL V34 HANDWRITING



```
3x=1
```

PRINTED REFERENCE: matched canvas / approximate ink footprint



PP-FormulaNet-S

EM: True | cells: None | render: rendered

$$3x = 1$$

```
3x=1
```

EM: True | cells: None | render: rendered

$$3x = 1$$

```
3x=1
```

PP-FormulaNet_plus-S

EM: True | cells: None | render: rendered

$$3x = 1$$

```
3x=1
```

EM: True | cells: None | render: rendered

$$3x = 1$$

```
3x=1
```

PP-FormulaNet-L

EM: True | cells: None | render: rendered

$$3x = 1$$

```
3x=1
```

EM: True | cells: None | render: rendered

$$3x = 1$$

```
3x=1
```

UniMERNet

EM: True | cells: None | render: rendered

$$3x = 1$$

```
3x=1
```

EM: True | cells: None | render: rendered

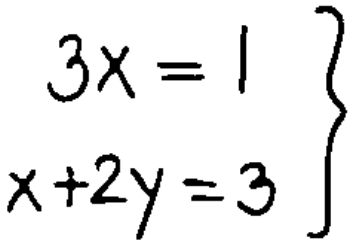
$$3x = 1$$

```
3x=1
```

Same GT and model. Printed/handwritten rendering geometry is approximate, not a perfectly isolated font intervention.

lecture_01_10423__formula_system

ACTUAL V34 HANDWRITING



```
\left.\begin{aligned}3x&=1\\x+2y&=3\end{aligned}\right\}
```

PRINTED REFERENCE: matched canvas / approximate ink footprint

$$\left.\begin{aligned}3x&=1\\x+2y&=3\end{aligned}\right\}$$

PP-FormulaNet-S

EM: False | cells: None | render: failed

```
3left+x=1\right\
```

EM: False | cells: None | render: rendered

$$\left.\begin{array}{c}{{3x=1}}\\ {{x+2y=3}}\end{array}\right\}$$

```
\left.\begin{array}{c}{{3x=1}}\\ {{x+2y=3}}\end{array}\right\}
```

PP-FormulaNet_plus-S

EM: False | cells: None | render: rendered

$$3x=3$$

```
3x=3
```

EM: False | cells: None | render: rendered

$$\left.\begin{array}{r}{3x=1}\\ {x+2y=3}\\ \end{array}\right\}$$

```
\left.\begin{array}{r}{3x=1}\\ {x+2y=3}\\ \end{array}\right\}
```

PP-FormulaNet-L

EM: False | cells: None | render: rendered

$$\{3x=1\}\{x+2y=3\}$$

```
\{3x=1\}\{x+2y=3\}
```

EM: False | cells: None | render: rendered

$$\left.\begin{aligned}3x&{{}=1}\\ {x+2y}&{{}=3}\end{aligned}\right\}$$

```
\left.\begin{aligned}{3x}&{{}=1}\\ {x+2y}&{{}=3}\end{aligned}\right\}
```

UniMERNet

EM: False | cells: None | render: rendered

$$x+2y=3$$

```
x+2y=3
```

EM: False | cells: None | render: rendered

$$\begin{array}{r}{3x=1\bigg\}}\\ {x+2y=3\bigg\}}\end{array}$$

```
\begin{array}{r}{3x=1\bigg\}}\\ {x+2y=3\bigg\}}\end{array}
```

Same GT and model. Printed/handwritten rendering geometry is approximate, not a perfectly isolated font intervention.

lecture_02_44601__formula

ACTUAL V34 HANDWRITING



x+3y=4

PRINTED REFERENCE: matched canvas / approximate ink footprint

$$x + 3y = 4$$

x+3y=4

PP-FormulaNet-S

EM: True | cells: None | render: rendered

$$x + 3y = 4$$

x+3y=4

EM: True | cells: None | render: rendered

$$x + 3y = 4$$

x+3y=4

PP-FormulaNet_plus-S

EM: True | cells: None | render: rendered

$$x + 3y = 4$$

x+3y=4

EM: True | cells: None | render: rendered

$$x + 3y = 4$$

x+3y=4

PP-FormulaNet-L

EM: True | cells: None | render: rendered

$$x + 3y = 4$$

x+3y=4

EM: True | cells: None | render: rendered

$$x + 3y = 4$$

x+3y=4

UniMERNet

EM: True | cells: None | render: rendered

$$x + 3y = 4$$

x+3y=4

EM: True | cells: None | render: rendered

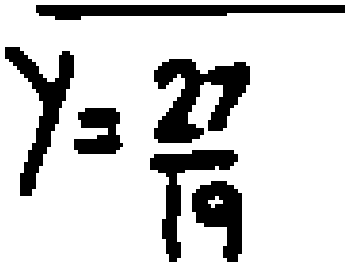
$$x + 3y = 4$$

x+3y=4

Same GT and model. Printed/handwritten rendering geometry is approximate, not a perfectly isolated font intervention.

lecture_02_44601__formula_fraction

ACTUAL V34 HANDWRITING



```
y=\frac{27}{19}
```

PRINTED REFERENCE: matched canvas / approximate ink footprint

$$y = \frac{27}{19}$$

PP-FormulaNet-S

EM: True | cells: None | render: rendered

$$y = \frac{27}{19}$$

```
y=\frac{27}{19}
```

EM: False | cells: None | render: rendered

$$y = \frac{27}{19}$$

```
y={\frac{27}{19}}
```

PP-FormulaNet_plus-S

EM: False | cells: None | render: rendered

$$y = \frac{2pi}{sqrtg}$$

```
y=\frac{2pi}{sqrt{g}}
```

EM: True | cells: None | render: rendered

$$y = \frac{27}{19}$$

```
y=\frac{27}{19}
```

PP-FormulaNet-L

EM: True | cells: None | render: rendered

$$y = \frac{27}{19}$$

```
y=\frac{27}{19}
```

EM: False | cells: None | render: rendered

$$y = \frac{27}{19}$$

```
y={\frac{27}{19}}
```

UniMERNet

EM: True | cells: None | render: rendered

$$y = \frac{27}{19}$$

```
y=\frac{27}{19}
```

EM: True | cells: None | render: rendered

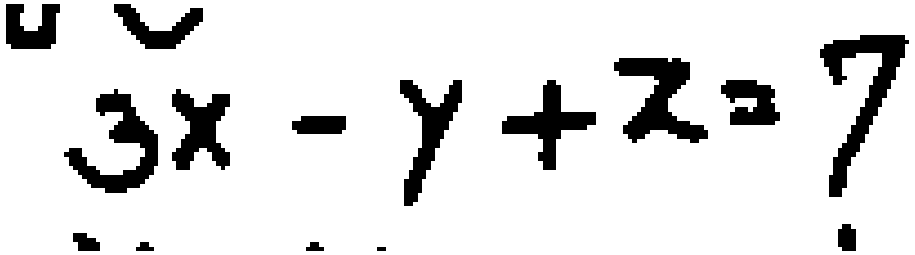
$$y = \frac{27}{19}$$

```
y=\frac{27}{19}
```

Same GT and model. Printed/handwritten rendering geometry is approximate, not a perfectly isolated font intervention.

nm_lecture_01_19899__formula

ACTUAL V34 HANDWRITING



```
3x-y+z=7
```

PRINTED REFERENCE: matched canvas / approximate ink footprint

$$3x - y + z = 7$$

PP-FormulaNet-S

EM: True | cells: None | render: rendered

$$3x - y + z = 7$$

```
3x-y+z=7
```

EM: True | cells: None | render: rendered

$$3x - y + z = 7$$

```
3x-y+z=7
```

PP-FormulaNet_plus-S

EM: False | cells: None | render: rendered

$$3x - y + x = 7$$

```
3x-y+x=7
```

EM: True | cells: None | render: rendered

$$3x - y + z = 7$$

```
3x-y+z=7
```

PP-FormulaNet-L

EM: True | cells: None | render: rendered

$$3x - y + z = 7$$

```
3x-y+z=7
```

EM: True | cells: None | render: rendered

$$3x - y + z = 7$$

```
3x-y+z=7
```

UniMERNet

EM: True | cells: None | render: rendered

$$3x - y + z = 7$$

```
3x-y+z=7
```

EM: True | cells: None | render: rendered

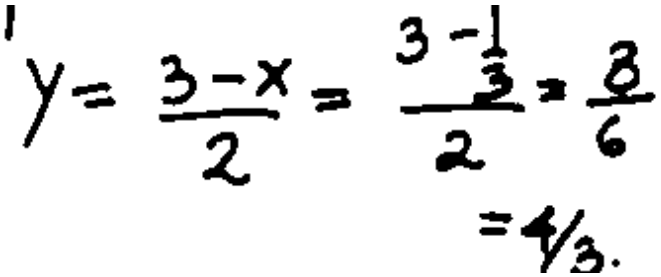
$$3x - y + z = 7$$

```
3x-y+z=7
```

Same GT and model. Printed/handwritten rendering geometry is approximate, not a perfectly isolated font intervention.

nm_lecture_01_19899__formula_fraction

ACTUAL V34 HANDWRITING



```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}
```

PRINTED REFERENCE: matched canvas / approximate ink footprint

$$y = \frac{3-x}{2} = \frac{3-\frac{1}{3}}{2} = \frac{8}{6}$$

PP-FormulaNet-S

EM: True | cells: None | render: rendered

$$y = \frac{3-x}{2} = \frac{3-\frac{1}{3}}{2} = \frac{8}{6}$$

```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}
```

EM: False | cells: None | render: rendered

$$y = \frac{3-x}{2} = \frac{3-\frac{1}{3}}{2} = \frac{8}{6}$$

```
y={\frac{3-x}{2}}={\frac{3-{\frac{1}{3}}}{2}}={\frac{8}{6}}
```

PP-FormulaNet_plus-S

EM: False | cells: None | render: rendered

$$y = \frac{3-x}{2} = \frac{3-\frac{1}{3}}{2} = \frac{3}{6}$$

```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{3}{6}
```

EM: True | cells: None | render: rendered

$$y = \frac{3-x}{2} = \frac{3-\frac{1}{3}}{2} = \frac{8}{6}$$

```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}
```

PP-FormulaNet-L

EM: False | cells: None | render: rendered

$$y = \frac{3-x}{2} = \frac{3-\frac{1}{3}}{2} = \frac{8}{6} = 4_{3.}$$

```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}=4_{3.}
```

EM: False | cells: None | render: rendered

$$y = \frac{3-x}{2} = \frac{3-\frac{1}{3}}{2} = \frac{8}{6}$$

```
y={\frac{3-x}{2}}={\frac{3-{\frac{1}{3}}}{2}}={\frac{8}{6}}
```

UniMERNet

EM: True | cells: None | render: rendered

$$y = \frac{3-x}{2} = \frac{3-\frac{1}{3}}{2} = \frac{8}{6}$$

```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}
```

EM: False | cells: None | render: rendered

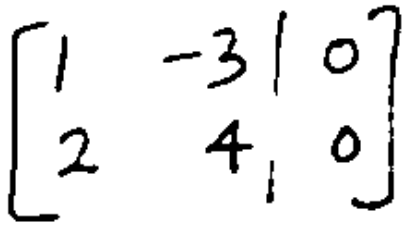
$$y = \frac{3-x}{2} = \frac{3-\frac{1}{3}}{2} = \frac{8}{6}$$

```
y={\frac{3-x}{2}}={\frac{3-{\frac{1}{3}}}{2}}={\frac{8}{6}}
```

Same GT and model. Printed/handwritten rendering geometry is approximate, not a perfectly isolated font intervention.

lecture_06_47799__augmented_matrix

ACTUAL V34 HANDWRITING



```
\left[\begin{array}{cc|c}1&-3&0\\2&4&0\end{array}\right]
```

PRINTED REFERENCE: matched canvas / approximate ink footprint

$$\left[\begin{array}{cc|c} 1 & -3 & 0 \\ 2 & 4 & 0 \end{array}\right]$$

PP-FormulaNet-S

EM: False | cells: False | render: rendered

$$[2, 310]]$$

```
[2,310]]
```

EM: False | cells: False | render: unsupported

```
\left[\begin{array}{c c|c}{1}&{-3}&{0}\\ {2}&{4}&{0}\ \ \ \
```

lecture_06_47799__real_space

ACTUAL V34 HANDWRITING



```
\mathbb{R}^{2}
```

PRINTED REFERENCE: matched canvas / approximate ink footprint



PP-FormulaNet-S

EM: False | cells: None | render: rendered

$$R^z$$

```
R^{z}
```

EM: True | cells: None | render: rendered

$$\mathbb{R}^2$$

```
\mathbb{R}^{2}
```

PP-FormulaNet_plus-S

EM: False | cells: None | render: rendered

$$P^z$$

```
P^{z}
```

EM: True | cells: None | render: rendered

$$\mathbb{R}^2$$

```
\mathbb{R}^{2}
```

PP-FormulaNet-L

EM: False | cells: None | render: rendered

$$R^2$$

```
R^{2}
```

EM: True | cells: None | render: rendered

$$\mathbb{R}^2$$

```
\mathbb{R}^{2}
```

UniMERNet

EM: False | cells: None | render: rendered

$$R^2$$

```
R^{2}
```

EM: True | cells: None | render: rendered

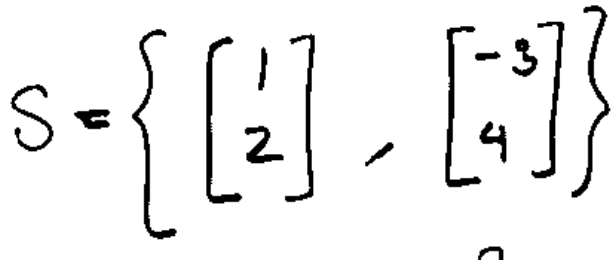
$$\mathbb{R}^2$$

```
\mathbb{R}^{2}
```

Same GT and model. Printed/handwritten rendering geometry is approximate, not a perfectly isolated font intervention.

lecture_06_47799__vector_set

ACTUAL V34 HANDWRITING



```
S=\left\{\begin{bmatrix}1\\2\end{bmatrix},\begin{bmatrix}-3\\4\end{bmatrix}\right\}
```

PP-FormulaNet-S

EM: False | cells: False | render: unsupported

```
S=\int\[\frac{1}{\sum2}]_{j}^{-1}]_delta{j}^{-}}\\\\ endend
```

PP-FormulaNet_plus-S

EM: False | cells: False | render: unsupported

```
s to int limits{{}}}_{{d}}sim{{}
```

PP-FormulaNet-L

EM: False | cells: False | render: unsupported

```
S=\{[1\atop2],[4]\}
```

UniMERNet

EM: False | cells: False | render: rendered

$$S = \{[\frac{1}{2}], [\frac{-3}{4}]\}$$

```
S=\{[\frac{1}{2}],[\frac{-3}{4}]\}
```

PRINTED REFERENCE: matched canvas / approximate ink footprint

$$S = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -3 \\ 4 \end{bmatrix} \right\}$$

EM: False | cells: False | render: unsupported

```
S=\left\{\left[\begin{matrix}{1}\\ {2}\\
\end{matrix}\right],\left[\begin{matrix}{-3}\\ {4}\ {\right{}}\right{\}}\end{array}
```

EM: False | cells: True | render: rendered

$$S = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -3 \\ 4 \end{bmatrix} \right\}$$

```
S=\left\{\left[\begin{matrix}{1}\\ {2}\\
\end{matrix}\right],\left[\begin{matrix}{-3}\\ {4}\\ \end{matrix}\right]\right\}
```

EM: False | cells: True | render: rendered

$$S = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -3 \\ 4 \end{bmatrix} \right\}$$

```
S=\left\{\left[\begin{matrix}{1}\\ {2}\end{matrix}\right],\left[\begin{matrix}{-3}\\
{4}\end{matrix}\right]\right\}
```

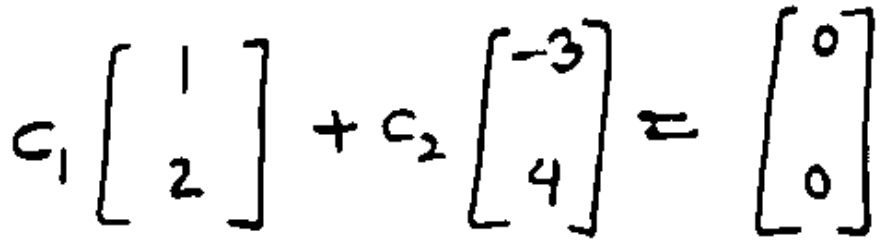
EM: False | cells: False | render: unsupported

```
S=\left\{{\textstyle\left[1\atop2\right]},{\textstyle\left[{-3\atop4}\right]}\right\}
```

Same GT and model. Printed/handwritten rendering geometry is approximate, not a perfectly isolated font intervention.

lecture_06_47799__linear_combination

ACTUAL V34 HANDWRITING



PRINTED REFERENCE: matched canvas / approximate ink footprint

$$c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

```
c_1\begin{bmatrix}1\\2\end{bmatrix}+c_2\begin{bmatrix}-3\\4\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}
```

PP-FormulaNet-S

EM: False | cells: False | render: rendered

$$c_1[\frac{1}{2}] + c_2[\frac{-1}{2}] = \sum 0limits_0^1$$

```
c_{1}[\frac{1}{2}]+c_{2}[\frac{-1}{2}]=\sum0limits_{0}^{1}
```

PP-FormulaNet_plus-S

EM: False | cells: False | render: unsupported

```
a_{1}[i\ frac{1}{z2}]+a_{2}[\frac{3}{z}]^{-3}[\]{2}[\frac{1}{z}]
```

PP-FormulaNet-L

EM: False | cells: False | render: rendered

$$c_1[2] + c_2[4] = [0]$$

```
c_{1}[2]+c_{2}[4]=[0]
```

UniMERNet

EM: False | cells: False | render: rendered

$$c_1[2] + c_2[3] = [0]$$

```
c_{1}[2]+c_{2}[3]=[0]
```

EM: False | cells: True | render: rendered

$$c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

```
c_{1}\left[\begin{matrix}{1}\\ {2}\\
\end{matrix}\right]+c_{2}\left[\begin{matrix}{-3}\\ {4}\\
[Full literal source in evaluation.json]
```

EM: False | cells: True | render: rendered

$$c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

```
c_{1}\left[\begin{matrix}{1}\\ {2}\\
\end{matrix}\right]+c_{2}\left[\begin{matrix}{-3}\\ {4}\\
[Full literal source in evaluation.json]
```

EM: False | cells: True | render: rendered

$$c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

```
c_{1}\left[\begin{matrix}{1}\\ {2}\\
\end{matrix}\right]+c_{2}\left[\begin{matrix}{-3}\\ {4}\\
[Full literal source in evaluation.json]
```

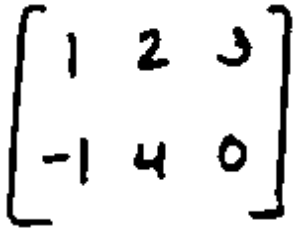
EM: False | cells: False | render: unsupported

```
c_{1}\left[1\atop2\right]+c_{2}\left[-3\atop4\right]=\left[0\atop0\right]
```

Same GT and model. Printed/handwritten rendering geometry is approximate, not a perfectly isolated font intervention.

lecture_07_67415__rectangular_matrix

ACTUAL V34 HANDWRITING



```
\begin{bmatrix}1&2&3\\-1&4&0\end{bmatrix}
```

PRINTED REFERENCE: matched canvas / approximate ink footprint

$$\begin{bmatrix} 1 & 2 & 3 \\ -1 & 4 & 0 \end{bmatrix}$$

PP-FormulaNet-S

EM: False | cells: False | render: rendered

$$[123]$$

```
[123]
```

EM: False | cells: True | render: rendered

$$\left[\begin{matrix}{1}&{2}&{3}\\ {-1}&{4}&{0}\\ \end{matrix}\right]$$

```
\left[\begin{matrix}{1}&{2}&{3}\\ {-1}&{4}&{0}\\ \end{matrix}\right]
```

PP-FormulaNet_plus-S

EM: False | cells: False | render: unsupported

```
[int\limits_{[1Object{]}}^{[]2}]^{T}}\\ {\ 1\ 4H}\\ \end{aligned}
```

EM: False | cells: True | render: rendered

$$\left[\begin{matrix}{1}&{2}&{3}\\ {-1}&{4}&{0}\\ \end{matrix}\right]$$

```
\left[\begin{matrix}{1}&{2}&{3}\\ {-1}&{4}&{0}\\ \end{matrix}\right]
```

PP-FormulaNet-L

EM: False | cells: False | render: rendered

$$[123]$$

```
[123]
```

EM: False | cells: True | render: rendered

$$\left[\begin{matrix}{1}&{2}&{3}\\ {-1}&{4}&{0}\\ \end{matrix}\right]$$

```
\left[\begin{matrix}{1}&{2}&{3}\\ {-1}&{4}&{0}\\ \end{matrix}\right]
```

UniMERNet

EM: False | cells: False | render: rendered

$$[-140]$$

```
[-140]
```

EM: False | cells: True | render: rendered

$$\left[\begin{array}{l l l}{1}&{2}&{3}\\ {-1}&{4}&{0}\end{array}\right]$$

```
\left[{\begin{array}{l l l}{1}&{2}&{3}\\ {-1}&{4}&{0}\end{array}}\right]
```

Same GT and model. Printed/handwritten rendering geometry is approximate, not a perfectly isolated font intervention.

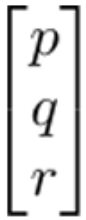
lecture_07_67415__symbolic_vector

ACTUAL V34 HANDWRITING



```
\begin{bmatrix}p\\q\\r\end{bmatrix}
```

PRINTED REFERENCE: matched canvas / approximate ink footprint



PP-FormulaNet-S

EM: False | cells: False | render: rendered

$$\int_{i}^{p}$$

```
\int\limits_{i}^{p}
```

EM: False | cells: True | render: rendered

$$\left[\begin{matrix}p\\ q\\ r\\ \end{matrix}\right]$$

```
\left[\begin{matrix}p\\ q\\ r\\ \end{matrix}\right]
```

PP-FormulaNet_plus-S

EM: False | cells: False | render: rendered

$$\int_{}^{g} intfracg\frac{g}{z}$$

```
\int\limits_{}^{g}int frac{{g}}{{\frac{g}{z}}}
```

EM: False | cells: True | render: rendered

$$\left[\begin{matrix}p\\ q\\ r\\ \end{matrix}\right]$$

```
\left[\begin{matrix}p\\ q\\ r\\ \end{matrix}\right]
```

PP-FormulaNet-L

EM: False | cells: False | render: rendered

$$[\frac{p}{q}]$$

```
[\frac{p}{q}]
```

EM: False | cells: True | render: rendered

$$\left[\begin{matrix}p\\ q\\ r\\ \end{matrix}\right]$$

```
\left[\begin{matrix}p\\ q\\ r\\ \end{matrix}\right]
```

UniMERNet

EM: False | cells: False | render: rendered

$$[\frac{p}{r}]$$

```
[\frac{p}{r}]
```

EM: False | cells: False | render: rendered

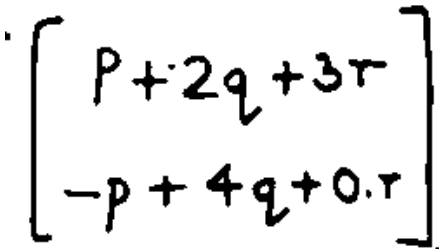
$$\binom{p}{q}$$

```
\binom{p}{q}
```

Same GT and model. Printed/handwritten rendering geometry is approximate, not a perfectly isolated font intervention.

lecture_07_67415__product_vector

ACTUAL V34 HANDWRITING



```
\begin{bmatrix}p+2q+3r\\-p+4q+0\cdot r\end{bmatrix}
```

PRINTED REFERENCE: matched canvas / approximate ink footprint

$$\begin{bmatrix} p+2q+3r \\ -p+4q+0\cdot r \end{bmatrix}$$

PP-FormulaNet-S

EM: False | cells: False | render: unsupported

```
\int\limits_{p+2q+3r}^{\ p+4q+0.r}^{\
```

EM: False | cells: True | render: rendered

$$\left[\begin{matrix}{p+2q+3r}\\ {-p+4q+0\cdot r}\\ \end{matrix}\right]$$

```
\left[\begin{matrix}{p+2q+3r}\\ {-p+4q+0\cdot r}\\ \end{matrix}\right]
```

PP-FormulaNet_plus-S

EM: False | cells: False | render: unsupported

```
[sqrt r geq2q+3\pi]^{-}]}{(-p+4q+0.r)}_{0}^{-1}
```

EM: False | cells: True | render: rendered

$$\left[\begin{matrix}{p+2q+3r}\\ {-p+4q+0\cdot r}\\ \end{matrix}\right]$$

```
\left[\begin{matrix}{p+2q+3r}\\ {-p+4q+0\cdot r}\\ \end{matrix}\right]
```

PP-FormulaNet-L

EM: False | cells: False | render: rendered

$$[\frac{p+2q+3r}{-p+4q+0.r}]$$

```
[\frac{p+2q+3r}{-p+4q+0.r}]
```

EM: False | cells: True | render: rendered

$$\left[\begin{matrix}{p+2q+3r}\\ {-p+4q+0\cdot r}\end{matrix}\right]$$

```
\left[\begin{matrix}{p+2q+3r}\\ {-p+4q+0\cdot r}\end{matrix}\right]
```

UniMERNet

EM: False | cells: False | render: rendered

$$[P+2q+3r]$$

```
[P+2q+3r]
```

EM: False | cells: True | render: rendered

$$\left[\begin{array}{c}{p+2q+3r}\\ {-p+4q+0\cdot r}\end{array}\right]$$

```
\left[\begin{array}{c}{p+2q+3r}\\ {-p+4q+0\cdot r}\end{array}\right]
```

Same GT and model. Printed/handwritten rendering geometry is approximate, not a perfectly isolated font intervention.

lecture_07_67415__noncommuting_product

ACTUAL V34 HANDWRITING



```
AB\ne BA.
```

PRINTED REFERENCE: matched canvas / approximate ink footprint



```
AB\ne BA.
```

PP-FormulaNet-S

EM: False | cells: None | render: rendered

$$AB \neq \beta A.$$

```
A B\neq\beta A.
```

EM: False | cells: None | render: rendered

$$AB \neq BA.$$

```
A B\neq B A.
```

PP-FormulaNet_plus-S

EM: False | cells: None | render: rendered

$$AB \neq BA.$$

```
A B\neq B A.
```

EM: False | cells: None | render: rendered

$$AB \neq BA.$$

```
A B\neq B A.
```

PP-FormulaNet-L

EM: False | cells: None | render: rendered

$$AB \neq BA.$$

```
A B\neq B A.
```

EM: False | cells: None | render: rendered

$$AB \neq BA.$$

```
A B\neq B A.
```

UniMERNet

EM: False | cells: None | render: rendered

$$AB \neq BA.$$

```
A B\neq B A.
```

EM: False | cells: None | render: rendered

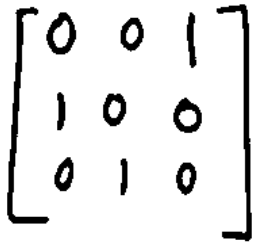
$$AB \neq BA.$$

```
A B\neq B A.
```

Same GT and model. Printed/handwritten rendering geometry is approximate, not a perfectly isolated font intervention.

lecture_08_53016__permutation_matrix

ACTUAL V34 HANDWRITING



```
\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}
```

PRINTED REFERENCE: matched canvas / approximate ink footprint

$$\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}$$

PP-FormulaNet-S

EM: False | cells: False | render: unsupported

```
\left.\begin{matrix}{0.01}\\ {0.00}\\ {0.0}\\ {0.0}\\ {end0}right\right
```

EM: False | cells: True | render: rendered

$$\left[\begin{matrix}{0}&{0}&{1}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\\ \end{matrix}\right]$$

```
\left[\begin{matrix}{0}&{0}&{1}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\\ \end{matrix}\right]
```

PP-FormulaNet_plus-S

EM: False | cells: False | render: rendered

$$\left[\begin{matrix}{0}&{1}\\ {0}&{0}\\ {0}&{0}\\ \end{matrix}\right]$$

```
\left[\begin{matrix}{0}&{1}\\ {0}&{0}\\ {0}&{0}\\ \end{matrix}\right]
```

EM: False | cells: True | render: rendered

$$\left[\begin{matrix}{0}&{0}&{1}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\\ \end{matrix}\right]$$

```
\left[\begin{matrix}{0}&{0}&{1}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\\ \end{matrix}\right]
```

PP-FormulaNet-L

EM: False | cells: False | render: rendered

$$\begin{matrix}001\\ [1000]\\ 010\end{matrix}$$

```
[\begin{matrix}{001}\\ {1000}\\ {010}\\ \end{matrix}]
```

EM: False | cells: True | render: rendered

$$\left[\begin{matrix}{0}&{0}&{1}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\\ \end{matrix}\right]$$

```
\left[\begin{matrix}{0}&{0}&{1}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\\ \end{matrix}\right]
```

UniMERNet

EM: False | cells: False | render: failed

```
\left[\begin{array}{l l}{0}&{0}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\end{array}\right]
```

EM: False | cells: True | render: rendered

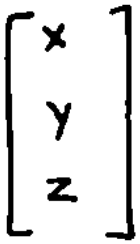
$$\left[\begin{array}{l l l}{0}&{0}&{1}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\end{array}\right]$$

```
\left[\begin{array}{l l l}{0}&{0}&{1}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\end{array}\right]
```

Same GT and model. Printed/handwritten rendering geometry is approximate, not a perfectly isolated font intervention.

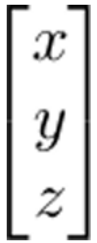
lecture_08_53016__input_vector

ACTUAL V34 HANDWRITING



```
\begin{bmatrix}x\\y\\z\end{bmatrix}
```

PRINTED REFERENCE: matched canvas / approximate ink footprint



```
\begin{bmatrix}x\\y\\z\end{bmatrix}
```

PP-FormulaNet-S

EM: False | cells: False | render: unsupported

```
\left[\^{x}\right]_{z}
```

EM: False | cells: True | render: rendered

$$\left[\begin{matrix}{x}\\ {y}\\ {z}\\ \end{matrix}\right]$$

```
\left[\begin{matrix}{x}\\ {y}\\ {z}\\ \end{matrix}\right]
```

PP-FormulaNet_plus-S

EM: False | cells: True | render: rendered

$$\left[\begin{matrix}{x}\\ {y}\\ {z}\\ \end{matrix}\right]$$

```
\left[\begin{matrix}{x}\\ {y}\\ {z}\\ \end{matrix}\right]
```

EM: False | cells: True | render: rendered

$$\left[\begin{matrix}{x}\\ {y}\\ {z}\\ \end{matrix}\right]$$

```
\left[\begin{matrix}{x}\\ {y}\\ {z}\\ \end{matrix}\right]
```

PP-FormulaNet-L

EM: False | cells: True | render: rendered

$$\left[\begin{array}{l}{x}\\ {y}\\ {z}\end{array}\right]$$

```
\left[\begin{array}{l}{x}\\ {y}\\ {z}\end{array}\right]
```

EM: False | cells: True | render: rendered

$$\left[\begin{matrix}{x}\\ {y}\\ {z}\end{matrix}\right]$$

```
\left[{\begin{matrix}{x}\\ {y}\\ {z}\end{matrix}}\right]
```

UniMERNet

EM: False | cells: True | render: rendered

$$\begin{array}{l}{x}\\ {y}\\ {z}\end{array}$$

```
[\begin{array}{l}{x}\\ {y}\\ {z}\end{array}]
```

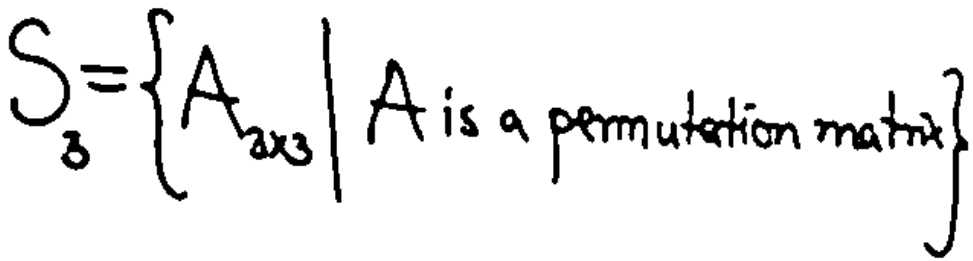
EM: False | cells: False | render: unsupported

```
\left[{x\atop y}\right]
```

Same GT and model. Printed/handwritten rendering geometry is approximate, not a perfectly isolated font intervention.

lecture_08_53016__set_builder

ACTUAL V34 HANDWRITING



PRINTED REFERENCE: matched canvas / approximate ink footprint

$$S_3 = \{A_{3\times3} \mid A \text{ is a permutation matrix}\}$$

```
S_3=\left\{A_{3\times3}\mid A\text{ is a permutation matrix}\right\}
```

PP-FormulaNet-S

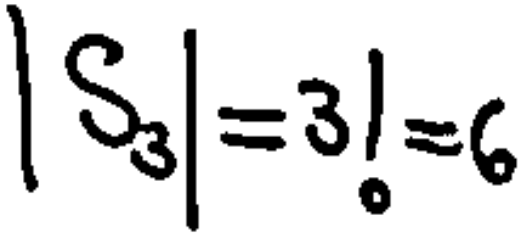
EM: False | cells: None | render: rendered

$$S_3 = \{ \lim_{A3 \to A \to B \sin\ \ \{!!\}} \ \ !\ !!\ !;!;$$

```
S_{3}=\{\lim\limits_{A3\rightarrow A\rightarrow B\sin\ \ {!!}}\ \ !\ !!\ !;!;\;\;\;\;
\;\;\;\;
```

lecture_08_53016__cardinality_factorial

ACTUAL V34 HANDWRITING



```
|S_3|=3!=6
```

PRINTED REFERENCE: matched canvas / approximate ink footprint

$$|S_3| = 3! = 6$$

PP-FormulaNet-S

EM: False | cells: None | render: failed

```
|vert__{3}\vert=3_{1}=6
```

EM: False | cells: None | render: rendered

$$|S_3| = 3! = 6$$

```
|S_{3}|=3!=6
```

PP-FormulaNet_plus-S

EM: False | cells: None | render: rendered

$$|S_3| = 31 = G$$

```
|S_{3}|=31=G
```

EM: False | cells: None | render: rendered

$$|S_3| = 3! = 6$$

```
|S_{3}|=3!=6
```

PP-FormulaNet-L

EM: False | cells: None | render: rendered

$$|S_3| = 31 = 6$$

```
|S_{3}|=31=6
```

EM: False | cells: None | render: rendered

$$|S_3| = 3! = 6$$

```
|S_{3}|=3!=6
```

UniMERNet

EM: False | cells: None | render: rendered

$$|s_3| = 3| = 6$$

```
\vert s_{3}\vert=3\vert=6
```

EM: False | cells: None | render: rendered

$$|S_3| = 3! = 6$$

```
|S_{3}|=3!=6
```

Same GT and model. Printed/handwritten rendering geometry is approximate, not a perfectly isolated font intervention.